

# Independent particles and Hartree-Fock

W. Ryssens



# Summary of last time

- 1 Operators in second quantisation
- 2 One-body density matrix  $\rho \leftrightarrow$  Slater determinants  $|\Psi\rangle$
- 3 Reference vacua and quasi-particle operators
- 4 Normal ordering
- 5 Elementary contractions
- 6 The (extended) Wick theorem
- 7 Rewriting the nuclear Hamiltonian with Wick's theorem

## Independent particle states

# Rewriting the Hamiltonian

After choosing a reference vacuum  $|\Phi\rangle$  with associated  $\{a, a^\dagger\}$ :

$$\begin{aligned} H = T + V &= \sum_{\alpha\beta} t_{\alpha\beta} a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} a_\alpha^\dagger a_\beta a_\delta^\dagger a_\gamma \\ &= \langle\Phi|H|\Phi\rangle + \sum_{\alpha\beta} h_{\alpha\beta}[\rho] :a_\alpha^\dagger a_\beta: + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} :a_\alpha^\dagger a_\beta a_\delta^\dagger a_\gamma: \\ &= H_0 + H_1 \end{aligned}$$

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How can we find the spectrum/the ground state of this Hamiltonian?

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Actual implementations exploit all possible properties of the matrices!

# Hamiltonian partitioning

Introduce **auxiliary diagonal and normal-ordered** one-body operator

$$\bar{h} = \sum_a \bar{\epsilon}_a : \hat{a}_a^\dagger \hat{a}_a : \quad \text{with } \epsilon_1 \leq \epsilon_2 \leq \dots$$

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non-diagonal 1-body operator

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diagonal one-body operator

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$$H_1 = H_1^{1b} + H_1^{2b} = \sum_{\alpha\beta} [h_{\alpha\beta} - \epsilon_{\alpha} \delta_{\alpha\beta}] : a_{\alpha}^{\dagger} a_{\beta} : + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} : a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma} :$$

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## Independent particle modelling: neglect $H_1$

- Nucleon transfer  $\rightarrow$  many excitations are particle-hole-like
- We can hope to find  $\epsilon_{\alpha}$  such that  $h_{\alpha} - \epsilon_{\alpha} \delta_{\alpha\beta}$  is "small"



# Independent-particle picture

$$H_0 = T + \bar{U} = \sum_{i=1}^N \left( \frac{\vec{p}_i^2}{2m} + \bar{U}(\vec{r}_i) \right)$$

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## Eigenstates in $\mathcal{H}_1$

- Nucleon orbitals  $|\alpha\rangle = |nljm\tau\rangle$

$$H_0 |nljm\tau\rangle = e_{nlj\tau} |nljm\tau\rangle$$

such that

$$H_0 = \sum_{nljm\tau} e_{nlj\tau} b_{nljm\tau}^+ b_{nljm\tau}$$

- A shell is  $2j+1$ -fold degenerate
- Fill shells for solutions in  $\mathcal{H}_N$   
up to the Fermi energy!

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Q for next time: can you prove this for 1p-1h excitations?

Q for next time: can you prove what the eigenstates in  $\mathcal{H}_{N\pm 1}$  look like?

# Many-body eigenstates of $H_0$ in $\mathcal{H}_N$

- Excitation energies

$$\epsilon_{ij\cdots}^{ab\cdots} - \langle \Phi | H | \Phi \rangle = (\epsilon_a + \epsilon_b + \cdots) - (\epsilon_i + \epsilon_j + \cdots) \geq 0$$

such that  $|\Phi\rangle$  is the ground state of  $H_0$

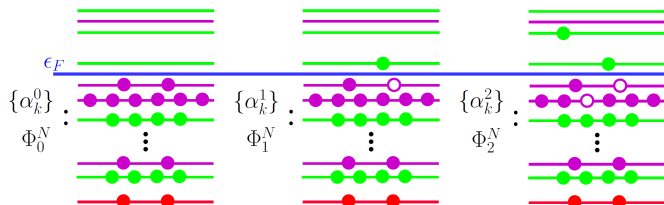
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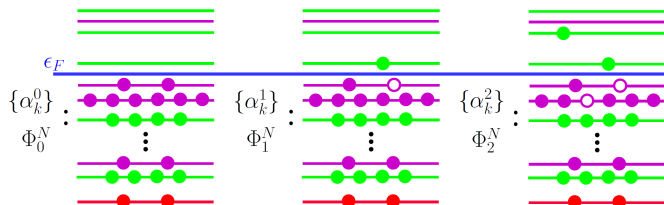
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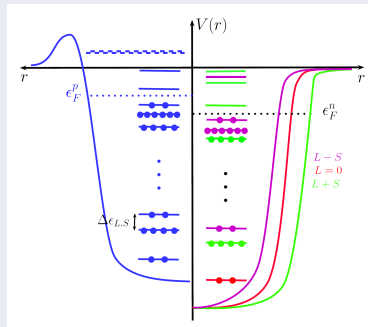
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- Excitation energy can be read from graphical representation

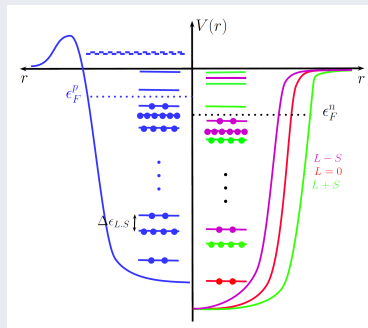
## Optimal $\bar{U}$ in $H_0^{1b}$ = Woods-Saxon



- Short range
- Attractive central component
- Flat bottom
- Coulomb effect for protons
- **Strong** spin-orbit component

- ✓ doubly closed-shell nuclei
  - One-nucleon transfer exp.
  - Magic numbers
  - Elements of spectroscopy
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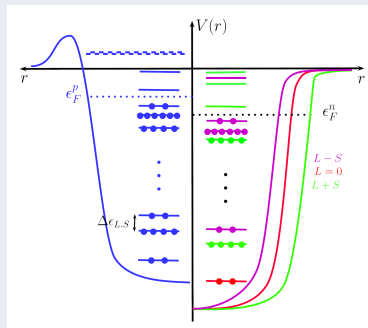


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# Discussion

- ✓ doubly closed-shell nuclei
  - One-nucleon transfer exp.
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  - ...
- ✓ Provides crucial insight
- ✗ Does not work universally
- ✗ Requires local adjustment
- ✗ Not all observables/nuclei
- ✗ Limited accuracy
- ✗ Unrelated to nuclear forces

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# Hartree-Fock

# The Hartree-Fock approximation

## The N-body problem

We want to solve

$$H|\Phi_n^N\rangle \equiv [H_0 + H_1]|\Phi_n^N\rangle = E|\Phi_n^N\rangle \text{ for } n = 1, 2, \dots$$

- $|\Phi_{ij\dots}^{ab\dots}\rangle \neq |\Phi_n^N\rangle$  if  $H_1 \neq 0$
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- How to choose the  $b^\dagger$  in an **optimal** way?
- What does 'optimal' even mean in this context?

# Ritz variational principle

We will determine the optimal  $b^\dagger$  by minimizing

$$E_0^{N(\text{HF})} \equiv \frac{\langle \Phi | H | \Phi \rangle}{\langle \Phi | \Phi \rangle} .$$

Notes:

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- 'Optimal' Slater determinant from the point of view of the energy.
- We will find an upper bound for the energy, i.e.  $E_0^{N(\text{HF})} \geq E_0^N$ ,  
since  $\{ \text{Slater determinants} \} \subset \mathcal{H}_N$ .

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Solving the Hartree-Fock problem is an optimisation problem:

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Reminder from lecture 2

- A Slater determinant  $\leftrightarrow$  its one-body density matrix  $\rho$ .
- The expectation value of the energy can be written as a function of  $\rho$ .

$$E[\rho] = \langle \Phi | H | \Phi \rangle = \sum_{\alpha\beta} t_{\alpha\beta} \rho_{\beta\alpha} + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \bar{v}_{\alpha\beta\gamma\delta} \rho_{\gamma\alpha} \rho_{\delta\beta}$$



# Minimising the energy

We demand that the energy be stationary under small variations:

$$\delta \left[ E[\rho] - \text{Tr} \left\{ \Lambda (\rho^2 - \rho) \right\} \right] = 0.$$

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Some work - see Sec. 6.3.2 - leads to

$$[h[\rho], \rho] = 0.$$

with **single-particle hamiltonian** or **Hartree-Fock mean-field**  $h$

$$h_{\nu\mu}[\rho] \equiv \frac{\partial E}{\partial \rho_{\mu\nu}} = t_{\nu\mu} + \sum_{\beta\delta} \bar{v}_{\nu\beta\mu\delta} \rho_{\delta\beta}.$$

 Functional derivative, see app. F

# Equations of motion

$$[h[\rho], \rho] = 0 \ .$$

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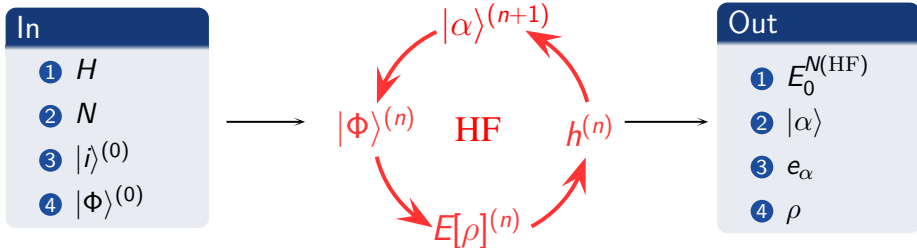
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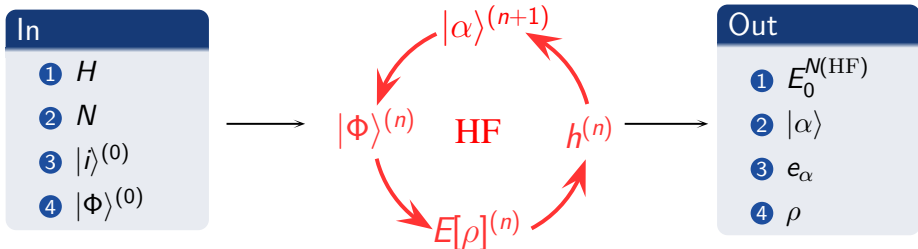
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- ! This problem is not easy to solve:  $h$  depends on  $\rho$ .

# Iterative procedure



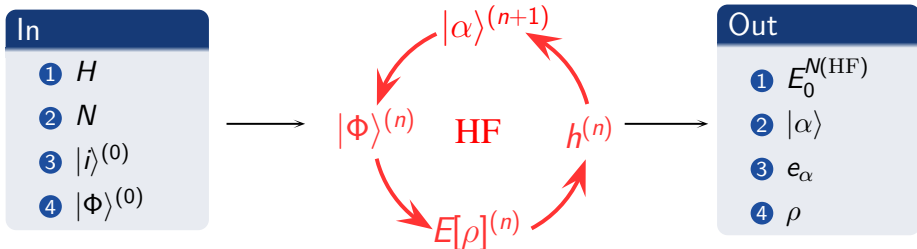


# Iterative procedure



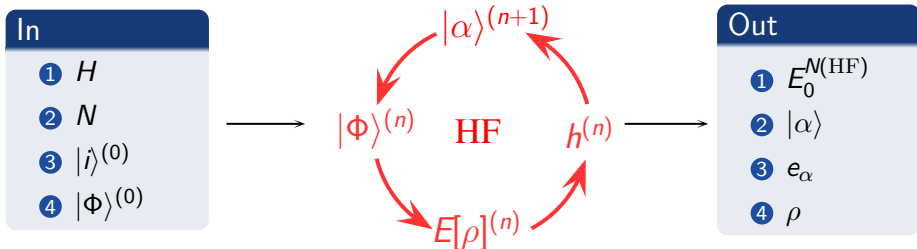
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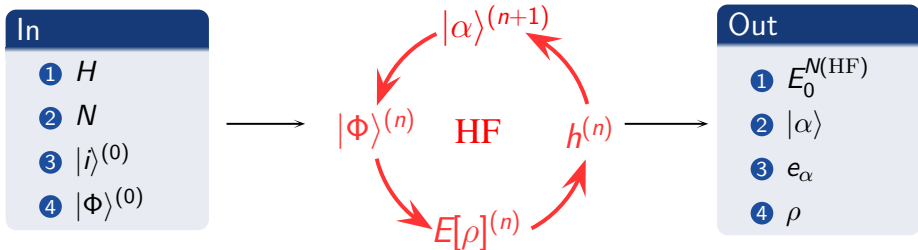
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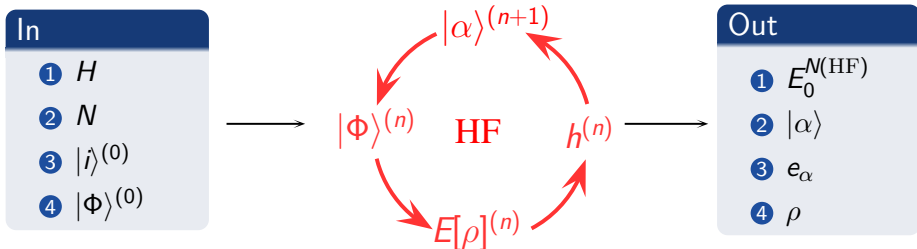
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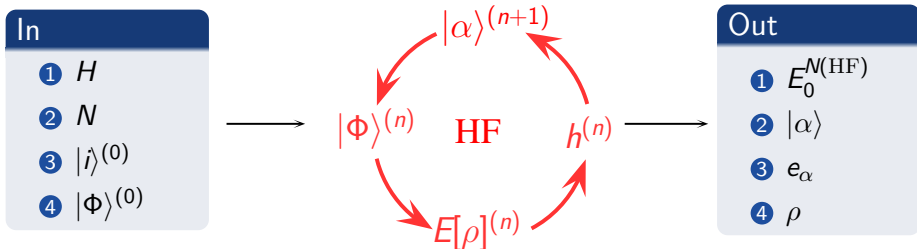
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- ! This never works naively - one has to work to get this to converge ...

# Coordinate-space representation

The energy (for a local interaction  $v(\vec{r}_1 - \vec{r}_2)$  and no spin or isospin)

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Fock or exchange potential: no classical analog

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- Advantageous in finite truncated basis  $\mathcal{B}_1$ : matrix diagonalization

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Q: what would be the energy of a non-interacting set of particles?

# Discussion: the Hamiltonian

Remember our partitioning of the Hamiltonian:

$$H_0 = H_0^{0b} + H_0^{1b} = \langle \Phi | H | \Phi \rangle + \sum_{\alpha} \epsilon_{\alpha} :a_{\alpha}^{\dagger} a_{\alpha}: .$$

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If  $|\Phi\rangle$  is the Hartree-Fock solution, then

- $H_1^{1b} = 0 \rightarrow$  the HF solution is "one-body optimal".
- In fact  $\langle \Phi | H | \Phi_i^a \rangle = 0 \rightarrow$  **Brillouins** theorem, see exercice session 2

# Discussion: the HF spectrum

Calculating the addition/removal energies:

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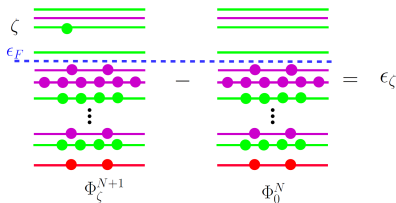


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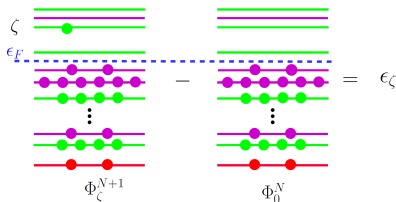


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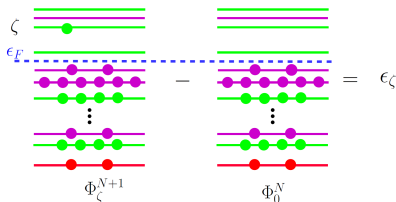
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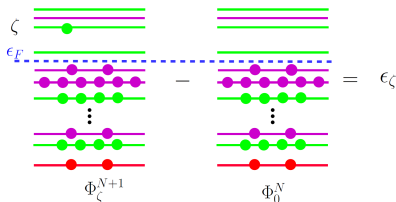
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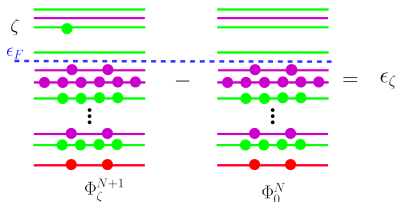
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- Bad approximation of  $E_k^{N\pm 1} - E_0^N$  if HF misses important correlations

# Discussion: the cost of a HF calculation

## Big O notation

$\mathcal{O}(f(N))$ : effort as function of input size  $n$  scales asymptotically as  $f(N)$ .

Example: diagonalising a matrix of dimension  $N$  is  $\mathcal{O}(N^3)$ .

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With  $N_b$  = the number of basis states:

- Exact solution:  $\mathcal{O}(X^3)$  with  $X = \begin{pmatrix} N_b \\ N \end{pmatrix}$

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Note:

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$$|\psi^{\text{HFB}}\rangle = \prod_{\mu} \beta |0\rangle = \text{quasi-particle product state}$$

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- MBPT: Many-body Perturbation Theory = at least  $\mathcal{O}(N^5)$

$$|\Psi^{\text{MBPT}}\rangle = |\Psi\rangle + \sum_{\alpha\beta ij} \frac{\langle\Psi_{ij}^{ab}|H_1|\Psi_{ij}^{ab}\rangle}{\epsilon_\alpha + \epsilon_\beta - \epsilon_i - \epsilon_j} |\Psi_{ij}^{ab}\rangle + \dots$$

# Going beyond Hartree-Fock

Note:

$$E^{N(\text{HF})} - E_0 = \Delta E^{N(\text{HF})} \neq 0$$

Other ansatzes:

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$$|\Psi^{\text{CC}}\rangle = \exp \mathcal{T} |\Psi^{\text{HF}}\rangle$$

$$\mathcal{T} = \sum_{ai} t_i^a a_a^\dagger a_i + \frac{1}{2} \sum_{\alpha\beta} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i + \dots$$

Singles cluster operator

Doubles

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Many strategies, with different strengths and generally high complexity.

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Can we do **better at HF level for  $\Delta E^{N(\text{HF})}$**  before invoking any of these?

Reminder: exercise session 2 will be online by the end of the week.